When More Measurements Cannot Help

A source-bound compiler for information-changing circuit experiments and conditional autonomous-service economics

Source-bound engineering study and conditional commercial execution

Evidence status. Exact finite-model measurement enablement; conventional comparison retained; physical assurance and the complete commercial flywheel remain unestablished.

1. Question, contribution and evidence boundary

Can an executable experiment compiler recognize that a measurement family cannot distinguish a consequential model class, construct a different observable, and retain the verified construction for another circuit? This study executes that question on two public ngspice passive-network designs. The first output is an experiment-design report, not a manufactured circuit, measured specimen, or customer order. The broader objective is to turn verified instruments into means for further discovery and production; the present evidence supports only part of that chain.

The implemented capability parses a bounded linear passive netlist, derives a scaling transformation from typed constitutive laws, produces exact responses and ambiguity witnesses, extends a voltage-only strategy grammar with input-admittance measurements, and selects a finite separating plan. A separate checker recomputes circuit behavior without importing candidate arithmetic. The retained instrument is tested on a different RLC topology under the same explicitly limited model semantics.

The primary cohort has 48 authored parameter hypotheses and 1,128 unordered pairs. The voltage-only arm separates 1,071; the constructed arm separates all 1,128. The second topology has 24 hypotheses and 276 pairs, with voltage-only separation of 252 and constructed separation of 276. Ordinary exhaustive experiment design attains the same complete coverage in each case. Its selected design is smaller than the initial greedy design. The added information matters; superior invention or search performance is not established.

All numerical cases are exposed DEVELOPMENT. Exact finite engineering checks are not statistical confirmation, and no protected scientific holdout or additional campaign error allocation is claimed. Candidate and verifier roles share a host, model family, tools and libraries. Their separation is procedural, not independent external certification. Public netlist provenance does not turn simulated responses into measured physical data.

A separate provider-identified current-frontier construction arm and a timed competent-human engineering study were not executed. Program reset and equal-content reuse are matched software controls; they are not new frontier API calls or measured expert labor. The study therefore does not establish superiority over independently operating frontier-AI construction, human scientific discovery or their best complete workflows.

Claim	Decision at this edition
Finite-model measurement enablement	Demonstrated within the frozen response and uncertainty model
Advantage over complete ordinary experiment design	Not demonstrated; complete coverage and action count tie
Scientific novelty or new circuit physics	Not established
Physical calibration and real diagnostic service assurance	Not validated
Actual commerce or autonomous frontier API execution	Not observed or authorized
Completed earned-funded invention flywheel	Not established; see separate financial and inheritance results

Claim decisions are narrower than the program mission. The complete connected commercial claim is not inferred from this table.

2. Source intake, cohort and acceptance obligation

The primary source is ngspice examples/measure/rc-meas-ac.sp at commit `032b1c32c4dbad45ff132bcfac1dbecadbd8abb0`, Git blob `2d464ba58ee46f4f88cb319debe5e90d396e06ad`. Its nominal network has two resistors and two capacitors. The second source is examples/Monte_Carlo/MonteCarlo.sp at the same source commit, Git blob `608dc4cfdda5d168952a6319d87a9d63f4f6cb84`. The latter contains an eight-element RLC network and an ngspice control block. The study extracts and models the nominal component section; it does not execute the source Monte Carlo procedure. Original files and the ngspice copying notice are preserved.

The admitted grammar supports positive finite R, C and L literals, one grounded unit AC voltage source, a bounded node and element count, and a limited set of directives. Active devices, parameter expressions, unknown source or load impedance and distributed or nonlinear behavior are outside this construction. Unmodeled effects do not receive zero uncertainty by default. Symbolic or unsupported inputs must be refused rather than silently approximated.

For the primary cohort, each of four nominal element values receives either 0.9 or 1.1 of its source value, then each resulting network receives impedance scale a in {1/2,1,2}. This produces 16 times 3= 48 hypotheses. The successor varies three selected elements, producing 8 times 3= 24 hypotheses. These are authored finite grids, not continuous tolerance intervals or a sample of physical manufacturing prevalence. The full hypothesis and pair denominators remain in every arm, including the stressed cases that cannot satisfy acceptance.

Quantity	**Primary RC**	**Second RLC**
Varied components	r1,c1,c2,r2	c1,c2,l3
Angular frequencies (rad/s)	100000; 300000; 1000000; 3000000	300000; 1000000; 3000000; 10000000
Hypotheses	48	24
Unordered pair denominator	1128	276
Voltage coordinate error	1/10000 in V/V	Same authored bound
Admittance coordinate error	1/10000000 S	Same authored bound
Admittance sensor gain	[999/1000,1001/1000]	Same authored interval

Protocol quantities are model assumptions. No instrument calibration establishes these error or gain bounds in a physical laboratory.

A delivery is accepted in the finite model only if every unordered pair has at least one admitted action whose complex-response uncertainty rectangles are disjoint. The endpoint concerns identification across the entire cohort, not mean error, a favorable subgroup, or the number of actions alone. Touching rectangles are not separated. Unseparated pairs are emitted explicitly and the complete identification claim abstains. The strong complete comparator receives the same netlist, model values, candidate actions and uncertainty policy.

$$\forall \{i,j\} \subset H, \quad \exists a \in A_{\text{selected}} : \; B_a(h_i) \cap B_a(h_j) = \varnothing.$$

The error rectangles are conservative coordinate-wise enclosures. Pairwise non-overlap is a sufficient finite identification certificate. Failure of this test is not, in general, proof that no inference method using joint nuisance structure could identify a pair. The global scale/gain counterexample below is stronger: it is an exact model equivalence. This distinction prevents a conservative verifier from becoming an unsupported impossibility theorem.

3. Constructed means: symmetry to a measurement extension

For a passive branch at positive angular frequency omega, the admittances are 1/R, j omega C and 1/(j omega L). The developer supplies these constitutive laws and their units. The program solves the exponent condition that every branch admittance scale by the same factor 1/a. It therefore emits R->aR, L->aL and C->C/a, together with coverage of the parsed branches and applicability exclusions. The algebraic derivation is executed; neither the physical laws nor the source selection are claimed to have been discovered autonomously.

$$R \mapsto aR, \qquad L \mapsto aL, \qquad C \mapsto C/a \quad \Longrightarrow \quad y_e \mapsto y_e/a.$$

At fixed normalized source voltage, the complete nodal matrix and forcing vector both scale by 1/a. When the nodal system is nonsingular, its voltage solution is unchanged at every node and every frequency. Consequently,

more voltage samples from this same model cannot identify global impedance scale. Input current, and hence input admittance, scales by 1/a. An admitted current observation changes the available information rather than merely searching harder over voltage reconstructions.

$$Yv = b \longmapsto (Y/a)v = b/a,$$
$$V_n/V_{\text{in}} \text{ unchanged}, \qquad I_{\text{in}}/V_{\text{in}} \mapsto (I_{\text{in}}/V_{\text{in}})/a.$$

The implementation uses exact rational complex arithmetic and Gaussian elimination, avoiding floating-point equality as evidence of structural non-identifiability. The initial voltage-only toolbox cannot express an input-admittance action. A generated extension adds that observable with its physical unit, frequency, acquisition-unit weight, derived symmetry and ancestry. A greedy finite strategy builder then chooses actions that separate unresolved pairs. The selected plan retains exact response tables, uncertainty policies, unresolved witnesses and source identities.

```
parse source netlist and validate bounded passive grammar
derive common-admittance scaling from typed branch laws
construct a distinct voltage-equivalent parameter witness
if old observation grammar preserves the witness:
generate an input-admittance measurement extension
select separating actions and emit exact pairwise certificates
independently recompute complete cohort coverage
```

This is a deliberately small executable representation extension. It is not unrestricted program synthesis, general autonomous experimental invention or a proof that a fixed menu has been transcended in arbitrary domains. Relative acquisition units rank actions inside the authored experiment model; they are not measured laboratory time or money. The initial greedy procedure is a constructive heuristic, with exact exhaustive set design available as a stronger ordinary alternative.

A distinct diffraction probe supplies a useful structural analogue. Two noncongruent six-point configurations have matching autocorrelation and therefore matching uncoded intensity at all frequencies; a known-type amplitude mask separates the exhibited pair. The relation transferred to circuits is equivalence under an observation family and separation by an information-changing intervention. No intensity-to-voltage numerical conversion is attempted. The typed correspondence search executes on representations extracted from checked witnesses; developers still choose the ontology and sources. The existing lifecycle strategy compiler refuses the unfamiliar refinement operator, and that refusal is retained. Equal-action ordinary construction reaches the same circuit result without the transferred hint, so the mapping is not evidence of discovery superiority.

4. Executed comparison and consequential counterexample

Complete-cohort endpoint	**Primary RC**	**Second RLC**
Voltage-only separated pairs	1,071/1,128	252/276
Constructed extended plan	1,128/1,128	276/276
Ordinary exhaustive extended plan	1,128/1,128	276/276
Constructed selected actions	1	1
Ordinary selected actions	1	1
Unknown-gain unresolved pairs	56	24

Executed selected-edition results; exact artifact identity is in the input manifest. Coverage changes follow the available observable. Complete ordinary experiment design remains a strong tie or improvement, not a defeated baseline.

The first r1 report was not accepted as a complete causal account: its per-step ancestry named unresolved scale witnesses that a selected voltage action could not distinguish, even though that action separated other pairs. The independent checker reproduced seven misleading ancestry claims. Version r2 corrected blocking-pair versus actually resolved-pair attribution under the unchanged cohort, frequencies, endpoint and uncertainty policy, and received 119/119 independent checks. Its reverse-delete policy still used two actions while exhaustive ordinary design used one; this adverse result is retained. A later policy repair must be judged only from its own actual receipt, rather than transferred from a favorable owner report.

Version r3 adds a conventional finite replacement check across the complete unchanged action universe after greedy selection and reverse deletion. An action is eligible as a replacement only if it alone covers every pair already achieved by the plan, with a lower admitted acquisition cost. The actual r3 report uses one action on both topologies and receives 131/131 checks from the frozen separate verifier. This closes the measured efficiency gap to ordinary exhaustive design; it does not outperform that control or establish a new algorithm. Every rejected replacement and its missing-pair evidence remains in the ancestry.

The candidate producer recorded 1.423969 process CPU seconds, 1.423026 wall seconds and 14,592 KiB maximum resident set size for both topologies and their comparison/stress arms. The separate trusted supervisor recorded 1.245008 waited-child CPU seconds and 0.01968618 supervisor CPU seconds for the original r1 run, with no surviving process-group descendants. These numbers are execution receipts under the stated host and source, not complete campaign costs, generalized performance benchmarks, or a monetary bill.

The most consequential reproduced defect is sensor-gain confounding. If measured current is g times physical current, replacing the circuit by its scale-a equivalent and replacing g by ag leaves every measured current and voltage response unchanged. A current channel without a justified calibration bound therefore does not establish global scale. The stressed gain interval[1/2,2] leaves unresolved pairs in both complete cohorts. The construction preserves those failures instead of shrinking the denominator or treating simulation settings as calibration evidence.

$$I_a = I_1/a, \qquad g_a = a g_1 \quad \Longrightarrow \quad g_a I_a = g_1 I_1.$$

The independent checker has its own parser, rational complex operations and nodal solver, and does not import candidate arithmetic. Closed-form RC and RL cases, source-amplitude normalization, SPICE M versus MEG units, touching uncertainty rectangles, large-error abstention, unsupported devices and singular networks provide adversarial checks. Its exact symmetry receipt covers the official RC design across three scales and five positive angular frequencies. The final full-cohort receipt and its immutable source hash are included in this edition source manifest. A passing unit suite alone is not substituted for complete-cohort artifact acceptance.

An additional review dispatch requested the gpt-6-astra alias at high reasoning effort and returned actual executable probes, while exact served revision and token billing remained unexposed. Its scalar-gain geometry check confirms that response rectangles can be conservative for individual actions; on the full declared action families it retains the same 56 and 24 unresolved pairs. The same review executes 34944 additional raw-reference corner cases, including negative-real and quadrature gains, with no analytic-enclosure escape. These are adversarial development probes, not a separate full frontier-model construction arm, outside certification, statistical confirmation or a provider-billed execution receipt.

Strong ordinary impedance spectroscopy already uses voltage and current. The experiment does not demonstrate an outcome beyond that practice. Its useful scope is an executable explanation of why a restricted observation family fails, an explicit repair to the measurement specification, and a checkable model-domain plan. It does not prove identifiability of arbitrary continuous component values, unknown circuit topology, nonlinear faults, phase-calibration errors or physical specimens.

5. Autonomous offering and complete-cost economics

The proposed narrow offering is a circuit measurement-design report: source intake, an exact ambiguity witness, a separating acquisition plan where one exists, uncertainty and calibration assumptions, and an independently checked report. The commercial comparison is autonomous frontier-AI-assisted delivery against competent human engineers supplying the same report and lifecycle obligations. Technical parity with conventional software does not invalidate a lower complete service cost. Conversely, a model-domain technical pass does not establish a buyer, price, service time or profitable enterprise.

Primary sources establish a broad electronics-services market and compensation sensitivities, not orders for this exact service. BLS reports electrical-engineer median annual pay of US$ 120630 for 2025. Using 2080 hours per year and the broad full-time private-industry total-compensation/wage ratio 54/36.97 gives an estimated loaded compensation sensitivity near US$ 84.71 per hour. This is not occupation-specific overhead or measured report effort. Provider consulting offers and marketplace listings are outside-option evidence; they are not AGI ALPHA willingness to pay. The source files retain currencies, dates and qualifiers. Canadian quotes are not silently converted into a USD account.

The current provider model documentation identifies gpt-6-astra and a standard tariff of US$ 10 per million input tokens and US$ 50 per million output tokens below its long-context threshold, with separate cached-input and

long-context terms. These are current availability and price observations. This campaign makes no newly paid API calls. The served revision, billable tokens and marginal bill for the inherited conversational orchestration are not exposed; therefore no API execution receipt or complete inference-cost measurement is claimed. A tariff multiplied by invented token counts would not close this gap.

$$\begin{aligned}\Pi_{\mathrm{AI}}(n) &= n\left[P - F(P) - C_{\mathrm{delivery}} - C_{\mathrm{risk}}\right] \\ &\quad - C_{\mathrm{setup}} - C_{\mathrm{research}}, \\ \Pi_{\mathrm{H}}(n) &= n\left[P - F(P) - w_{\mathrm{loaded}} t_{\mathrm{H}} - C_{\mathrm{tools}} - C_{\mathrm{risk,H}}\right] \\ &\quad - C_{\mathrm{setup,H}}.\end{aligned}$$

Task-matched human time remains unmeasured, so a human-to-AI profitability crossing is not established. The correct output is a frontier in human hours, missing autonomous cost, price and participation. If both routes are profitable under a given assumption set, the result is a cost or margin advantage rather than a claim that human work was uneconomic. The non-entry option, free self-service and competing AI-native providers remain relevant to investment attractiveness even when a human cost advantage is possible.

The simulated exchange reuses the existing pinned ProtectedExchange implementation. Its acceptance predicate binds the actual technical report and separate verifier receipt; its wallets, buyer preferences, funds and settlement authority are simulated. A payment-gated artifact can protect against deliberate nonpayment, while captured card payments remain reversible. Matching tariff assumptions include fixed and percentage fees, disputes and any countering cost. Reserves, refundable deposits and borrowing are not unrestricted operating earnings. End-of-horizon liabilities must be run off or retained before an investment is treated as funded.

Counterfactual world	**Participates**	**Profit (SIM USD)**	**Terminal cash (SIM USD)**
conditional participation	yes	12.25	1012.25
zero demand	no	-36.00	964.00
equal ai native outside option	no	-36.00	964.00
strategic nonpayment	yes	-36.00	964.00
rejected delivery	yes	-37.75	962.25
outage delay	yes	-37.75	962.25
interruption recovery	yes	12.25	1012.25
correlated benefit warranty reversals	yes	-87.75	912.25

One admitted report opportunity in each alternative world. Participation means modeled utility preference; funding or delivery can still fail. These are not additive sales, measured buyer frequencies or independent scientific samples. All admitted preparation cost remains when output is unsold.

In the nominal authored world, selling price is SIM USD 50, the declared all-category production, setup and research recovery cost is 36 before the 1.75 payment fee, and complete assumed cost is 37.75. This yields modeled profit 12.25 and terminal cash 1012.25 from declared opening capital 1000. Full modeled principal and dispute-fee exposure remains backed separately; a predeclared 10 investment reservation is not a spend or successor construction. These amounts are scenario inputs and ledger consequences, not measurements of the campaign bill.

The nominal missing-cost frontier is K<12.25 SIM USD above the assumed complete totals. At the sourced loaded-compensation sensitivity, human profit is positive before unmeasured tools, setup and service overhead only below 29.217221 minutes per report; the authored AI cost is below human compensation only above 20.540592 minutes. Exact human effort is unknown, so neither a measured human loss nor a measured AI cost advantage follows. Actual complete profit remains unbounded by the available bill and obligation evidence.

The evidence-grounded account records 0 real orders, 0.00 actual receipts and 0.00 qualified unrestricted simulated earned funds. The adapter decision is DEFERRED NO QUALIFIED EARNINGS. Conditional balances remain in the scenarios rather than silently funding a new real or evidence-qualified construction. The commercial run binds the exact accepted report identified in its authority receipt; subsequent technical policy versions do not retroactively change that delivered artifact.

Conditional positive scenario balances are retained as conditional results. They do not become evidence-qualified earned funds when unknown costs or independently ungrounded participation remain. The actual buyer-participation

frontier includes zero orders, low willingness to pay, switching and trust costs, and an equally capable AI-native outside option. Joint adverse scenarios must combine demand reduction, benefit overstatement, payment reversals and warranty exposure rather than varying each conveniently in isolation. No actual customer contact, payment or empirical commerce occurred.

6. Retention, successor construction and operational scope

A saved report alone is not an inheritance test. The retained path must store source identity, executable generative procedure, typed applicability, observed costs and failure boundaries; retrieve it against a fresh task; and actually invoke the inherited adaptation. This campaign uses the repository existing content-addressed retention API. The second topology is useful because it adds inductive branches and a different connection structure while retaining the limited passive one-port laws. It does not supply independent physical truth or an unexposed statistical confirmation set.

Policy	Toolbox	Pairs separated	Actions	Acquisition units
old-greedy	old-voltage	252/276	1	1
old-greedy	new-current	276/276	2	3
new-reverse-delete	old-voltage	252/276	1	1
new-reverse-delete	new-current	276/276	1	2

Each row represents three separately executed memory arms: retained, equal-content ordinary reuse and reset. The three arms tie on every displayed endpoint. The policy name new-reverse-delete also includes the later finite single-action replacement repair.

The actual retrieval sequence is flywheel.retention.store, flywheel.retention.retrieve, generate_actions(inherited_program), then compile_plan. The content-addressed instrument record is `6a793f083a9abd7076c0bd94b3afb6739f8264f29f0a778985796eebf3fa506c`. A separate incompatible unknown-gain query is refused by retained negative applicability knowledge. The program interprets a fixed typed data procedure, not arbitrary code from retrieved text. Source pinning and applicability checking execute before adaptation.

The twelve matched control arms recorded 2.529036 process CPU seconds and 2.531387 wall seconds in the producer receipt. Current-enabled plans separate all 276 pairs; old voltage-only plans leave 24 unresolved. The known new policy reduces authored acquisition units from 3 to 2 at unchanged complete coverage, while equal-content reuse, reset and strong ordinary design attain the same useful result. This isolates a toolbox-information effect and a known-policy efficiency repair; it does not demonstrate retained-memory discovery superiority, statistical learning or an after-cost economic gain.

Equal-content ordinary reuse receives the same verified instrument content; reset receives the complete permitted tools and may reconstruct the known impedance analysis. Old and new policy/toolbox combinations separate possession of an input-current observable from a superior discovery procedure. If inherited construction and ordinary reuse tie, the result is technical portability rather than retained-memory discovery advantage. Acquisition, adaptation, checking and failed searches remain part of lifecycle economics, even when marginal execution is fast.

6.1. A second construction from the verified gain defect

The gain-confounding witness was converted into a typed construction constraint and a paired-reference observation primitive. The new procedure measures normalized input current with a parallel reference conductance switched off, then switched on, using the same complex sensor transfer during both readings. It emits an executable paired-measurement DSL and a rational normalization expression. This changes the information available under stable but unknown gain. The reference technique is known ratiometry; no invention priority or superiority over the same-reference conventional control is claimed.

$$z_0 = gY + e_0, \qquad z_1 = g(Y + G) + e_1, \qquad \widehat{Y} = G_{\mathrm{nom}} \frac{z_0}{z_1 - z_0}.$$

In the noiseless exact-reference case, the shared complex transfer cancels algebraically. The executed protocol allows reference conductance between 0.99 and 1.01 times its nominal 0.001 S, common gain magnitude at least 0.5, and

coordinate noise at most 1e-10 S. These are authored conditional bounds, not laboratory calibration. The code derives an analytic conservative enclosure rather than treating the finite noise grid as a statistical confidence interval. A positive lower bound on the reference-difference denominator is required.

$$d = \gamma G_l - 4\epsilon > 0,$$
$$E = \frac{G_{\mathrm{nom}}}{G_l} \frac{2G_h\epsilon + 4\epsilon(|\Re Y| + |\Im Y|)}{d}.$$

Here G_l and G_h bound the known reference, gamma bounds the common transfer magnitude from below, and epsilon bounds each real and imaginary noise coordinate. The normalized estimate is enclosed around the reference-scale interval[Gnom/G_h,Gnom/G_l] times Y, enlarged by E in each coordinate. The proof uses conservative triangle-inequality bounds for the two complex errors. It remains conditional on the circuit/load model and the external stability and noise assertions.

Reference-extension endpoint	**Primary witness**	**New RLC family**
Admitted cases	2	24
Separated pairs	1/1	276/276
Selected frequencies	1	1
Ordinary reference-control frequencies	1	1
Raw readouts checked	600	7200
Within analytic conditional enclosure	600	7200
Unknown drift	Exact indistinguishability; abstain	Exact indistinguishability; abstain
High coordinate noise	Abstain	Abstain

The new RLC family varies r1,c3,l2 by 7/8 or 9/8 and then applies three impedance scales. These 24 newly authored model cases are exposed development, not protected statistical confirmation.

The separate reference checker passes 153/153 checks and independently recomputes every one of 7,800 raw paired readouts using its own parser, rational KCL implementation and bound derivation. The producer records 0.571674 process CPU seconds and 0.573056 wall seconds. Four exact drift witnesses for each topology preserve both raw readings across distinct networks when the transfer changes between the two measurements. The instrument therefore requires external stability evidence; it does not infer stable gain from the very two readings that depend on that assumption. A high-noise stress removes the positive denominator guarantee and produces abstention.

This additional capability was constructed from the original declared campaign allocation. Qualified unrestricted simulated earnings remained zero. The modeled ten-dollar investment reservation was not spent on this work and is not retrospectively described as its funding. The result demonstrates a causal conditional information change and a tested response to a verified criticism, while the financial compounding link and physical validation remain open.

Execution uses the existing trusted-command resource supervisor and declared limits, not a sandbox claim. CPU reservations are conservatively charged even where process accounting is incomplete. Cancellation and interruption checks must include startup and child cleanup. The historic frozen startup interruption failure from the preceding campaign is preserved separately; a green run here does not repair or erase it. Public source/evidence checkpoints and downloaded-content replay provide restart durability. Later paper or artifact verification does not count as a fresh scientific sample.

7. Opportunity decisions, limits and next discriminating work

Three materially different possibilities were investigated. Circuit measurement design was selected for bounded construction because a small public design permits an exact ambiguity witness, executable observation change and strong complete baseline. Diffraction supplied a real identifiability issue but known coded-acquisition methods already address the exhibited model; no hardware or calibrated experimental cohort was acquired. RNA isoform measurement supplied public lr-kallisto code/data references and a small real PAX2 graph, but that graph has no transcript paths and cannot support abundance inference; a manufacturer SIRV download was inaccessible. Existing structural-identifiability and long-read methods remain strong prior art. These observations prevent dataset convenience or an access failure from becoming a claim of a new scientific gap.

The first implemented tool can help expose when an experiment specification asks inference to recover information that its measurements do not contain. This function matters beyond one circuit only within its verified applicability: the common-admittance symmetry is topology-independent for the admitted complete passive network model, and the second topology checks a further source design. A broad market-size narrative is unnecessary and would not demonstrate demand. A civilization-scale invention claim would require consequential new work enabled beyond competent complete alternatives, with physical validity and economic consequence independently established.

The principal unresolved boundaries are physical model adequacy and calibration; unrestricted or continuous parameter identifiability; observed exact-service human effort; complete autonomous inference and enterprise research cost; AGI ALPHA buyer willingness to pay and repeatable volume; exposure to real settlement and warranty obligations; and after-cost discovery advantage over equal-content and competent AI-native alternatives. No failure is hidden by a new version name or a smaller confirmation cohort. No reported cash balance is treated as actual revenue.

The next decisive physical experiment is to apply the frozen report obligation to calibrated source-defined RC and RLC assemblies with independently varied impedance scale, known sensor error and phase behavior, and measured acquisition/review effort. The strongest complete voltage/current workflow must run on the same units. Commercial evidence then requires authorized empirical participation for the same report and liabilities. Those dependent actions are not authorized or provisioned by this study. Within the current authority, reproducible construction, counterexamples, conditional accounting and the exact restart instruction are the delivered checkpoint, with mission completion still withheld.

8. Reproduction and source identity

This edition is reconstructed from an immutable public checkpoint and the exact input identities in SOURCE_MANIFEST.json. Each consumed file is hashed before authoring; no absent historical array, unobserved bill or prior paper is reconstructed under an old identity. The report source anchor and input manifest distinguish the engineering source freeze from the paper derivative. The browser edition contains the actual narrative, equations, results and source links rather than an HTML listing of LaTeX code.

The 26 original raw-readout JSON files are preserved losslessly as named members of reference-raw-r1.tar.gz, rather than claimed to exist as separate public Git blobs. The manifest records their archive identity, exact member names, sizes and original SHA256 values. The paper builder checks those archived bytes directly, and checks any present local raw files for equality; no regenerated raw array inherits an old identity.

Public engineering anchor: `326be47e40df57f868b1543e4d5385d490319cea`. Repository: MontrealAI/AGI-Alpha-Invention-Flywheel-Lab. Source URL: https://github.com/MontrealAI/AGI-Alpha-Invention-Flywheel-Lab/tree/326be47e40df57f868b1543e4d5385d490319cea. The stable PR is https://github.com/MontrealAI/AGI-Alpha-Invention-Flywheel-Lab/pull/13. The input-manifest SHA256 is `b51eea9cb1bc8e3213c46af1b8ad6b6fdfda4aa389068e60f1967646dba44e35`.

```
python -m pip install -r requirements-consequential009.txt
python run_consequential009.py -output /tmp/consequential009-replay
python -m unittest discover -s verification/consequential009 -v
```

Use a fresh output directory. The initial complete-path attempt with an external output directory failed during verifier source-relative identity construction; its original failure and resource reservation are preserved. A narrow source/artifact namespace-label repair changes no numerical criterion, and a second actual outside-repository local replay completes in 7.373538 seconds with both independent acceptances. This is local source-matched replay; the historical result field naming a downloaded-source scope is not evidence that a download occurred. Actual downloaded-package and release-asset qualification require their own receipts. Do not overwrite or delete previous evidence to reuse an output name.

A fresh hosted qualification at the earlier d06ab792 checkpoint exposed an inherited runtime dependency on cryptography in the protected exchange. The circuit and checker arithmetic use the Python standard library; the complete exchange does not. The original failed check is retained. The current declared packaging edition pins cryptography 50.0.2, cffi 2.1.1 and pycparser 3.11 in requirements-consequential009.txt; fresh-environment and downloaded-package qualification are separate from the executed scientific results above. This typesetting/evidence annotation edition includes that explicit installation instruction without transferring an earlier pass to the changed head.

Re-execution can change timings while preserving exact mathematical claims. Downloaded-package qualification and source-matched CI are separate evidence scopes; consult the publication and verification receipts for their actual status at this edition. A test pass on a previous commit is not transferred automatically to a later head or release archive. Artifacts from reproduced runs are not counted as additional independent observations.

9. Primary sources and retained provenance

[1] Circuit sources and known AC analysis. https://github.com/ngspice/ngspice/tree/032b1c32c4dbad45ff132bcfac1dbecadbd8abb0/examples

[2] ngspice AC-analysis tutorial. https://ngspice.sourceforge.io/ngspice-tutorial.html

[3] OpenAI GPT-6 Astra model and tariff, accessed 30 October 2026. https://developers.openai.com/api/docs/models/gpt-6-astra

[4] BLS electrical and electronics engineers, May 2025 wage. https://www.bls.gov/ooh/architecture-and-engineering/electrical-and-electronics-engineers.htm

[5] BLS employer compensation, June 2026 release. https://www.bls.gov/news.release/ecec.nr0.htm

[6] Stripe US domestic-card and dispute tariff. https://stripe.com/pricing

[7] Fiverr broad electronics-services price summary, August 2026. https://www.fiverr.com/resources/guides/costs/electronics-design-engineer

[8] MB Designs Canadian consultation offer. https://mbdesignswork.com/

[9] Grimm and Baake, Homometric Point Sets and Inverse Problems, 2008. https://arxiv.org/abs/0808.0094

[10] Candes, Li and Soltanolkotabi, Phase Retrieval from Coded Diffraction Patterns. https://statistics.stanford.edu/technical-reports/phase-retrieval-coded-diffraction-patterns

[11] Isoform-deconvolution structural identifiability, 2022. https://academic.oup.com/bioinformatics/article/38/6/1491/6493231

[12] Long-read transcriptome quantification with lr-kallisto, 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12680354/

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